

AUTO 503 Project Report

Development and Validation of Suspension Model for Electric Passenger Vehicle

Submitted to:

Jeffrey Cook
Professor and Lecturer
Electrical Engineering & Computer Science Department
University of Michigan

Prepared by:

Mihir Tasgaonkar
MEng Automotive Engineering
University of Michigan

Date Submitted: 08/13/2024.

Contents

Contents	2
Executive Summary	3
1.0 Introduction and Current State.....	4
1.1 Overview of the Project	4
1.2 Review of Current State Design / Literature Review	5
1.3 Project Deliverables	6
1.4 Scope.....	6
1.5 Project Timing	6
2.0 Assumptions and Methodology	7
2.1 Project Methodology.....	7
2.2 Quarter Car Suspension System Dynamics	7
2.3 Free Body Diagram and Governing Equations.....	8
2.4 Simulink Model	9
3.0 Results and Discussion	11
3.1 Model Verification.....	11
3.2 Model Validation	13
4.0 Recommendations.....	14
5.0 Impact/Financial Benefits	14
6.0 Summary/Conclusions	15
7.0 References.....	15

Executive Summary

The automotive industry constantly seeks to enhance vehicle performance, comfort, and safety. One critical aspect is the suspension system, which directly impacts ride quality and handling. This project focuses on developing and validating a suspension model based on quarter car model equations. The project encompasses both theoretical development and practical validation, which will most likely be conducted in an automotive testing facility at the Lucid Motors Engineering Center in Newark, CA.

The major objectives for the project are as follows:

- Develop a robust suspension model based on quarter car model equations to simulate vertical dynamics accurately.
- Integrate the suspension model to bridge the gap between the vehicle body and wheels for comprehensive analysis.
- Tune model parameters to optimize ride comfort, handling, and stability.
- Validate the developed model using real-world data to ensure its accuracy and reliability.

Using the findings of this project, the sponsor company will have an enhanced understanding of the dynamic response of the vehicle, which will lead to improvements in suspension design as necessary. Furthermore, the development and validation of an accurate suspension model for simulation and analysis will help reduce development cost and lead to reduced time-to-market for the new models of the sponsor company. It will also enable optimization of dynamic parameters like ride comfort, handling, and stability, which translates into a competitive product on the market.

1.0 Introduction and Current State

1.1 Overview of the Project

Lucid Motors is an electric vehicle manufacturer with a luxury sedan, the Lucid Air, on the market, and a luxury SUV, the Lucid Gravity, about to be launched. The company is known for its high-performance powertrain systems that combines powerful electric motors with high energy-density lithium-ion batteries to provide an exciting driving experience as well as industry-leading figures of all-electric range. Lucid directly competes with other electric vehicle manufacturers such as Tesla and Fisker Inc.

The USP of the Lucid Motors vehicles is the focus on efficient powertrain design. A lot of importance is given to optimizing the overall packaging of the electric powertrain, thus giving a high-performance output while reducing the weight as much as possible. One of the essential components of this is the suspension design.

The aim of this project is to simulate and analyze the dynamic response of the suspension system theoretically and compare these results with the real-world data collected from vehicle testing. This process will be iterated multiple times to ensure that the suspension model developed for simulation is closely representative of the real vehicle while also not being computationally heavy. This will ensure accurate simulation results and faster iteration time, leading to an optimized development process. This accurate model can then be used by the algorithm development and validation teams within the company to add/improve the vehicle's dynamic features (such as electronic stability control, adaptive dampers, and adjustable ride height).

1.2 Review of Current State Design / Literature Review

There are three steps of control algorithm testing for automotive systems – model-in-loop (MIL), software-in-loop (SIL), and hardware-in-loop (HIL) testing. Each of these steps represents a crucial part of the software development V-cycle:

- MIL Testing – Verifying the logic of the control system works the way it intends to.
- SIL Testing – Designing and implementing test cases specifically for testing the robustness of the code and not the logic particularly.
- HIL Testing – Testing the interactions between the controller and its plant model in a simulated environment and diagnose communication issues.

In general, these three steps of software testing can be clubbed together as X-in-Loop testing. This will be referred to as XiL in this report henceforth.

Generally, the XiL model is used to configure the code for testing the various ECUs in the vehicle. For example, if the validation team wants to conduct HIL testing for the Vehicle Dynamics Controller (VDC), then the model can have the inputs and outputs for the VDC configured for HIL testing, while all other ECUs are configured for MIL (model-in-loop) testing, since their behavior needs to be in a simulated environment.

There are plant models present to simulate the components, such as high-voltage batteries, motors, drivetrain components, and the chassis components (vehicle body, suspension, steering, wheels). Since there are 60 ECUs in the Lucid Motors vehicles, manually configuring the model based on the ECU to be tested is a difficult task. This model helps in automating the process of configuring the model correctly.

The development of this model for simulating the behavior of all ECUs and plant models is the major responsibility of the Plant Modeling team at Lucid Motors, which I am a part of. Currently, there is no subsystem within the XiL model that simulates the effect of the forces applied by the suspension on the vehicle body as well as the wheels. The vehicle body is mostly considering longitudinal dynamics (forward direction), while the wheels subsystem calculates the wheel speed from the vehicle speed and the longitudinal forces acting on the wheel. The objective of the suspension plant model is to bridge the gap between the Body and the Wheels subsystems and simulate the suspension dynamics of the system, and possibly integrate other features such as semi-active damping.

1.3 Project Deliverables

The following list entails the deliverables from the project:

1. Developed suspension model based on quarter car model equations.
2. Integrated model bridging the gap between vehicle body and wheels.
3. Tuned parameters optimizing ride comfort, handling, and stability.
4. Validation report comparing simulation results with real-world data and outlining improvements.

1.4 Scope

The scope of this project is to develop a suspension model that captures the vertical dynamics to improve the results of the HIL testing process and allow the controls development and validation teams to gain a better understanding of the overall design and make room for improvements and optimization.

1.5 Project Timing

The entire project will be completed over the course of a summer internship starting 3rd June 2024 and ending 23rd August 2024.

The tentative timeline for the project would be as follows:

1. Completion of initial suspension model development – 14th July 2024
2. Integration of model – 30th July 2024
3. Parameter tuning - 7th August 2024
4. Validation report - 11th August 2024

2.0 Assumptions and Methodology

2.1 Project Methodology

The project will adopt a systematic approach involving theoretical development, data collection, simulation, and validation:

1. Theoretical Development: Derive and refine the quarter car model equations to accurately represent suspension vertical dynamics.
2. Simulation: Implement the developed suspension model and replay real-world scenarios using sample inputs first and actual vehicle inputs later.
3. Validation: Compare simulation results with actual data to validate the model's accuracy and identify areas for improvement.
4. Iterative Improvement: Refine the suspension model based on validation results to enhance its predictive capability.

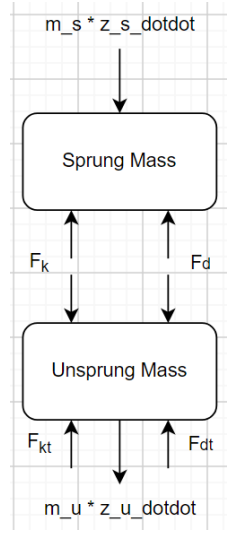
2.2 Quarter Car Suspension System Dynamics

A quarter car model is a simple representation of the vertical dynamics of the suspension system. It assumes the vertical dynamics of the system (response of the suspension system relative to the changes in the road profile in the vertical direction) is independent of the lateral and longitudinal dynamics of the vehicle body and wheels. However, it can be integrated with individual lateral and longitudinal models to simulate the vehicle body along all three axes.

This model consists of a double mass-spring-damper system. This means that there are two masses being considered – sprung mass and unsprung mass. The sprung mass is essentially all the vehicle mass that needs to be supported by the suspension system. The unsprung mass is the mass of all components within or below the suspension system (like springs, dampers, brakes, wheel hubs, etc.) This mass is supported by the tire.

The suspension system is represented by a spring and a damper, and a similar representation is used for the tire. The following section shows the free-body diagram for the same, as well as the governing equations for this kind of a system.

2.3 Free Body Diagram and Governing Equations



Based on the free body diagram above, the following governing equations can be used to model the suspension system:

$$m_s * \ddot{z}_s = -F_{sus} - F_{ext}$$

$$m_u * \ddot{z}_u = -F_{kt} - F_{dt} + F_{sus} + m_u * g$$

Where,

$$F_{sus} = F_k + F_d = \text{Suspension forces}$$

$$F_k = k_s \times (z_s - z_u) = \text{Suspension force due to spring}$$

$$F_d = b_s \times (\dot{z}_s - \dot{z}_u) = \text{Suspension force due to damper}$$

$$F_{kt} = k_t \times (z_u - z_r) = \text{Tire stiffness force}$$

$$F_{dt} = b_t \times \dot{z}_u = \text{Tire damping force}$$

$$m_s = \text{sprung mass (for a quarter car, it is the vehicle mass/4)}$$

$$m_u = \text{unsprung mass (mass of suspension, wheel, brakes, etc.)}$$

$$z_s = \text{displacement of sprung mass}$$

$$z_u = \text{displacement of unsprung mass}$$

$$k_s = \text{suspension spring stiffness}$$

$$b_s = \text{suspension damping coefficient}$$

$$k_t = \text{tire stiffness constant}$$

$$b_t = \text{tire damping coefficient}$$

$$z_r = \text{input road profile}$$

$$F_{ext} = \text{any other vertical loads acting on the vehicle (like aerodynamic downforce)}$$

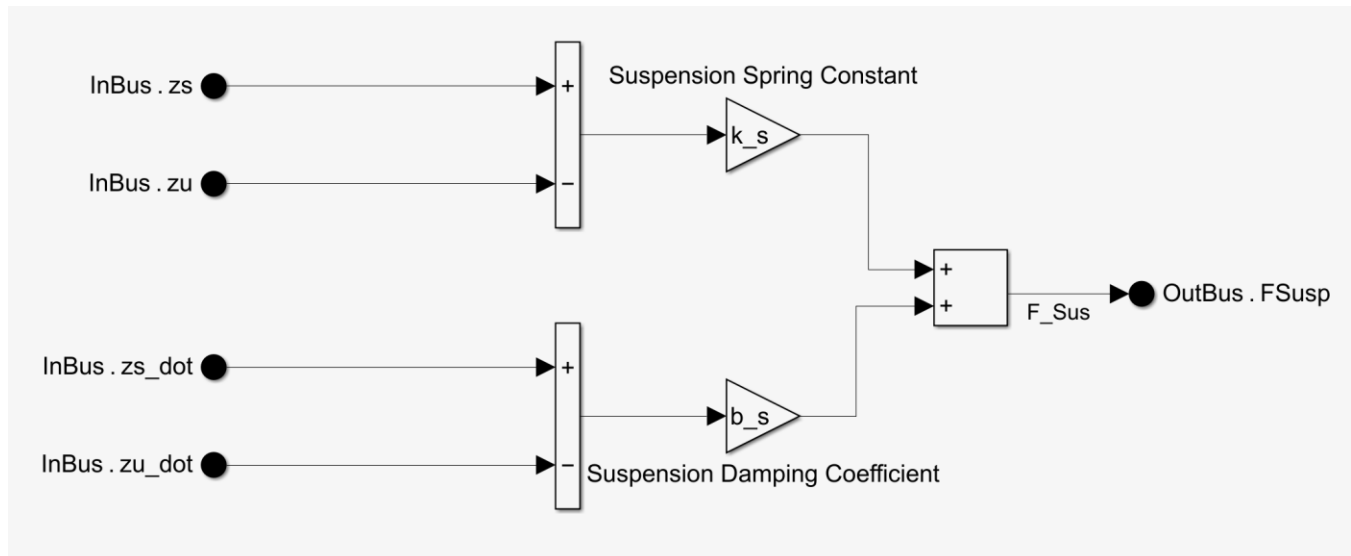
2.4 Simulink Model

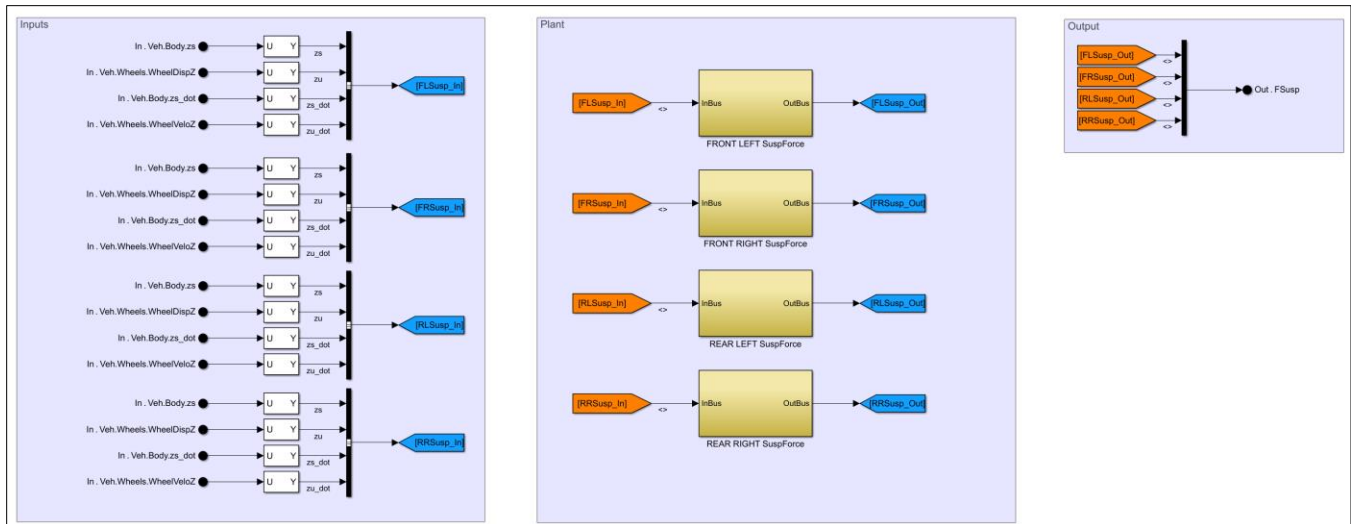
Now that the quarter car model has been represented in mathematical form, it can be modeled in Simulink. The model needs to be divided into three vehicle components:

- Vehicle body – The sprung mass calculations will be simulated here.
- Suspension – This consists of the suspension force calculations.
- Wheels – The unsprung mass displacement will be calculated in this subsystem.

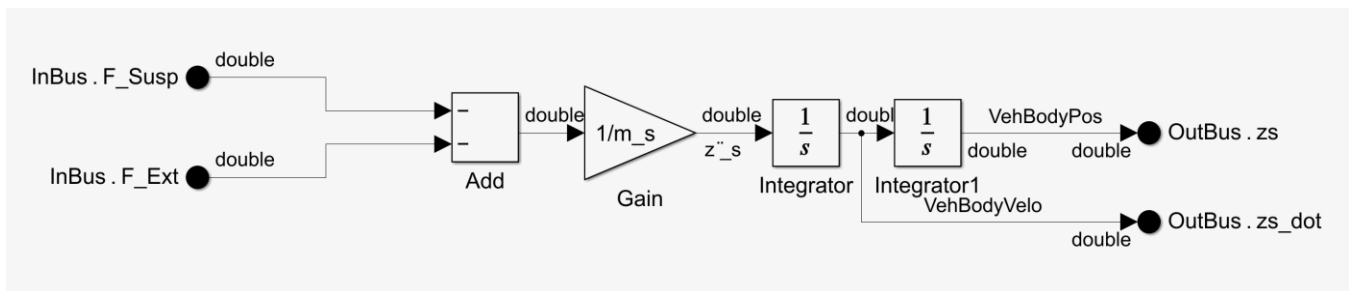
This segregation is based on the architecture of the XiL model. The full-vehicle model already contains the Wheels and the Vehicle Body subsystems, which currently simulate dynamics in the XY plane of the vehicle (longitudinal and lateral). Hence, the vertical displacement of the wheel (unsprung mass) as well as the vehicle body (sprung mass) are modeled within their respective subsystems, and the suspension forces are modeled within a new subsystem.

Suspension Subsystem

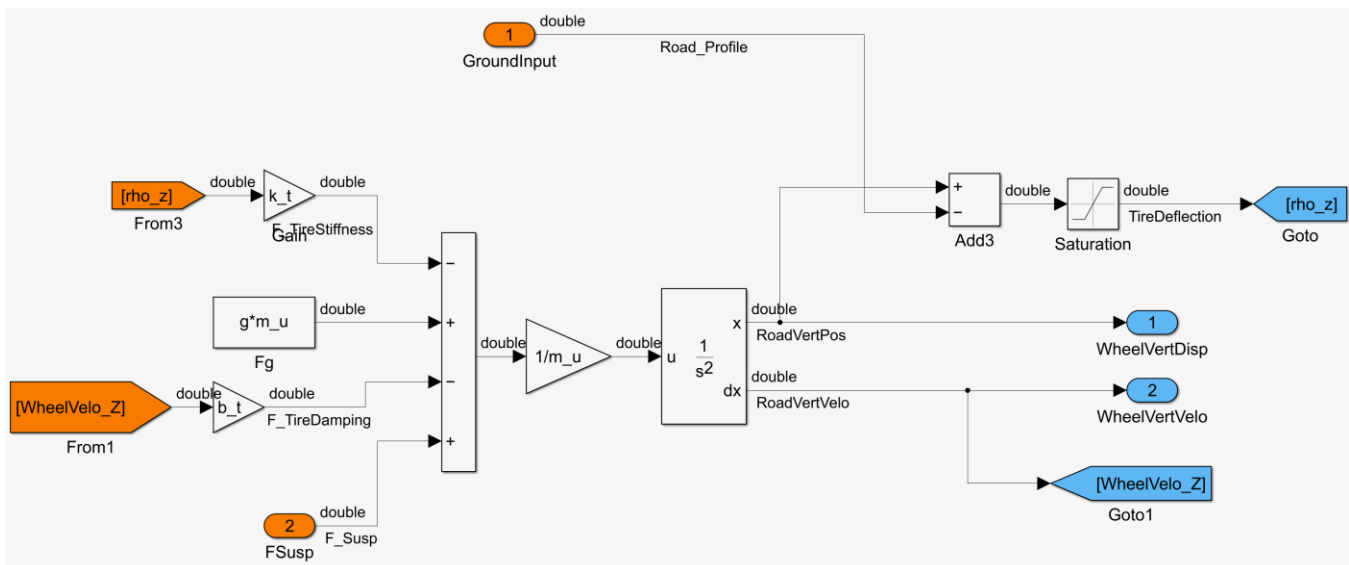




Vehicle Body Subsystem



Wheels Subsystem



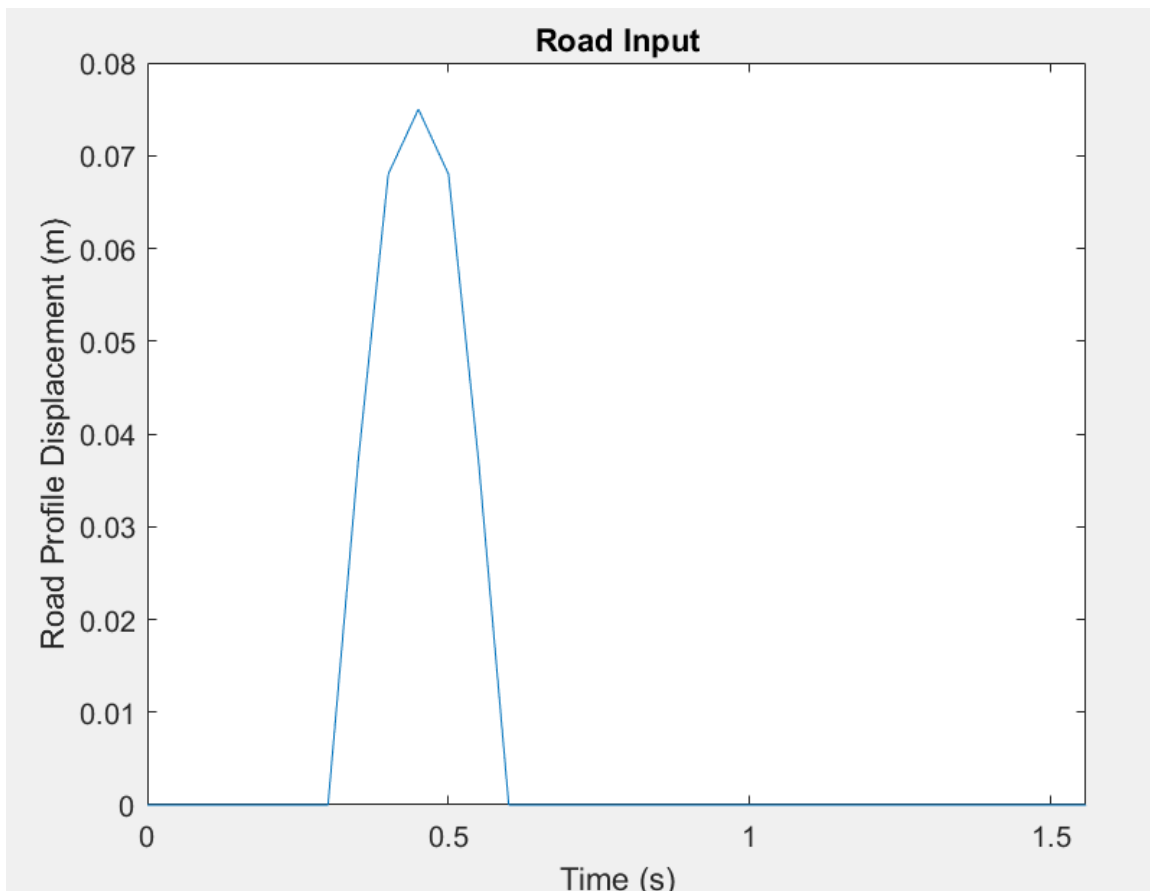
In the Wheels subsystem, the term ρ_z is used to define the difference between the unsprung mass displacement and road profile, which is essentially the tire deflection.

3.0 Results and Discussion

3.1 Model Verification

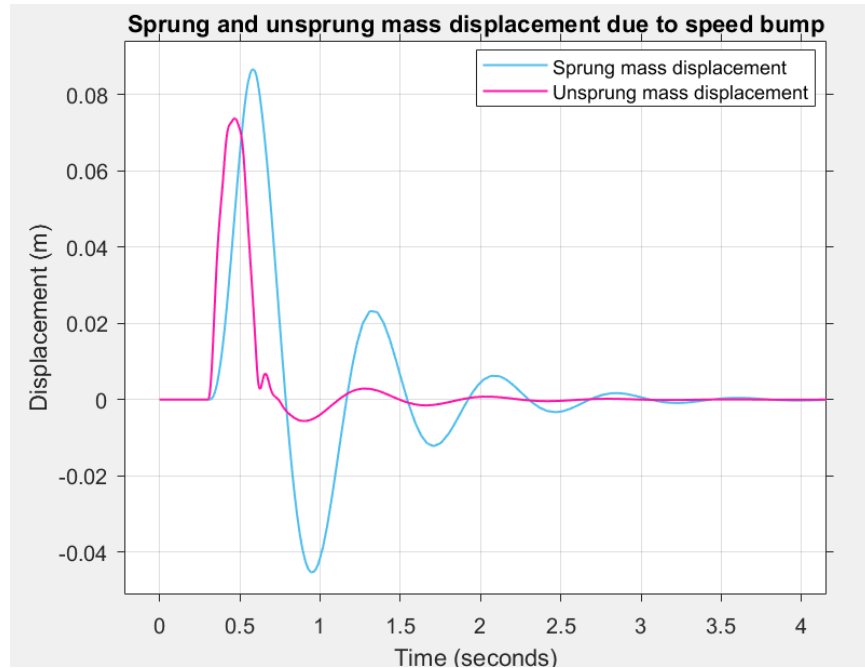
For the verification of the model, a model-in-loop was conducted. This type of testing involves creating a simple test rig with a sample road input to the model and analyzing the effect of this input on the sprung and unsprung mass displacements.

Since the suspension system in the Lucid vehicles uses a semi-active damper, there is no fixed value for the suspension damping coefficient (“ b_s ” in the model). Based on the damping curves in the suspension system, the damping coefficient can vary between values of 2473 N-s/m and 9438 N-s/m. The quarter car suspension model test rig was simulated for this range of damping coefficient values. The sample road input simulates a speed bump on the road and is shown in the graph below:



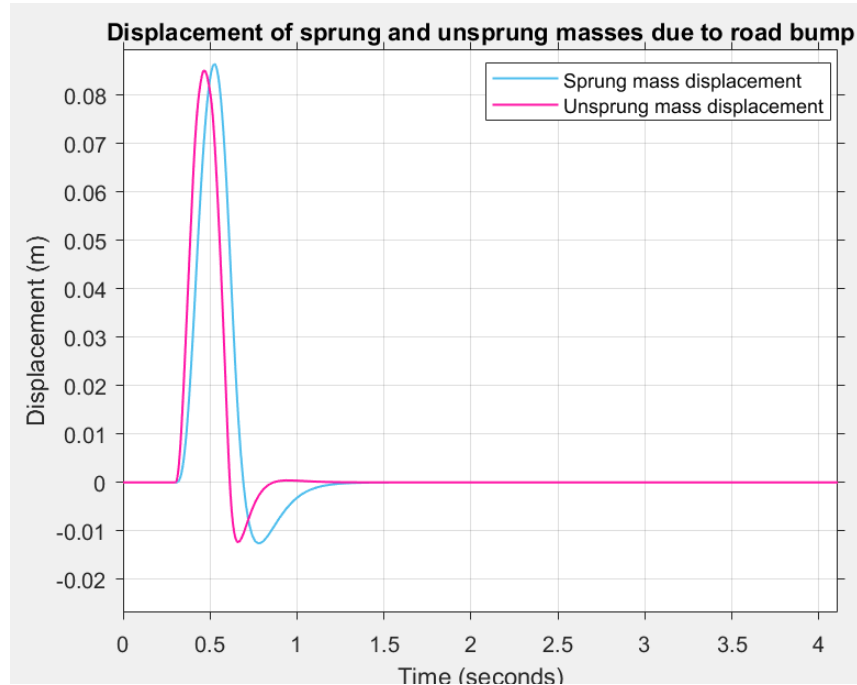
Road Profile simulating a speed bump between $t = 0.3$ to 0.6 sec

The following response is observed from the suspension subsystem based on the sprung mass displacement (movement of the vehicle body in the vertical direction) and the unsprung mass displacement (movement of the suspension system in the vertical direction):



Response of passive quarter car suspension model due to speed bump with suspension damping coefficient = 2473 N-s/m

The sprung mass settles to a steady-state value at approximately $t = 3.3$ sec, which is a settling time of roughly 3 seconds. Increasing the damping coefficient value reduces this settling time:



Response of passive quarter car suspension model due to speed bump with suspension damping coefficient = 9438 N-s/m

Here, the settling time is around 1 second due to the higher damping force acting on the vehicle body and the wheels by the suspension.

These results have been labeled as *satisfactory* and the model has thus been verified with a sample input.

3.2 Model Validation

For validating the model, there is no direct test data available regarding the sprung and unsprung mass displacements based on road profile inputs. However, the kinematics and compliance (K&C) test data as obtained from the vehicle dynamics testing team contains the static force values that the suspension exerts on the vehicle body for each wheel. The values of the suspension forces on each wheel after simulating the XiL model were compared with this test data and the table below shows the analysis:

Static Suspension Force (N)	Test Data	Simulation Data
Front Left Wheel	6973.97	7013.11
Front Right Wheel	7097.48	7013.11
Rear Left Wheel	6791.99	7387.97
Rear Right Wheel	6791.59	7387.97

The simulation test data assumes the external forces acting on the suspension from the vehicle body is same for the right and left tires on either axle, while the test data will also consider the deviations in weight because of lateral as well as longitudinal weight distribution.

The quarter car model has thus been validated based on the static forces in the simulation being close to what is seen in the test data in addition to the sprung and unsprung mass displacement responses being satisfactory as seen in the verification test.

4.0 Recommendations

Thus, using quarter car equations for simulations, the vertical dynamics of the vehicle's suspension system have been modeled and validated. While this is a straightforward way of modeling the suspension's vertical dynamics, it gives very little idea about the overall dynamics of the system. The effects of the vertical forces on the longitudinal and lateral dynamics of the vehicle cannot be captured using this kind of modeling technique. A potential next step for this project would be develop a model that captures these effects in over the full vehicle body.

Furthermore, the effects of the suspension geometry (toe, caster, camber) can also be integrated within this model to make it more accurate as well as provide additional flexibility of simulating multiple conditions and suspension setups.

5.0 Impact/Financial Benefits

While this model does not completely reveal the interactions between the vehicle body and wheels by itself, it is a good beginning to modeling the dynamics of the vehicle's suspension system. By implementing the recommendations discussed in the previous section, the dynamics of the vehicle body in all three places can be captured and the model can also allow for additional flexibility of simulation by integrating suspension geometry into the model.

Nonetheless, the working of the suspension system was verified based on a sample input, and the suspension force calculations were validated against test data collected through K&C testing to make sure the model is on the right track.

Since this model is intended to be a small part of a much larger simulation effort, it is hard to directly quantify the impact of the model on the organization. However, the cost of operating a Hardware-in-Loop test bench can be as much as \$1 per minute, so there is a large focus on improving the accuracy of simulation models to avoid testing model-level intricacies on the HIL bench. By making the simulation model as accurate as possible, the HIL test bench can be kept free only for testing hardware-specific communications, and the XIL model that my team is working on is expected to help that effort.

HIL testing is also a highly time-consuming process. Compiling the models to be ready for a HIL test itself can take up to 4-5 hours, so every effort of reducing the amount of HIL tests within the process without impacting the accuracy and the output of the organization will result in major time savings.

6.0 Summary/Conclusions

The objective of this project was to kick off the modeling and simulation of the suspension system within the XIL testing process within the organization. This initial model was to be developed using quarter car suspension equations, since it helps in decentralizing the four wheels to individual systems and understand their dynamics in an isolated manner.

Using quarter car equations, a suspension model was developed and integrated with the existing full-vehicle body. The model also went through a well-defined verification and validation process, where the settling time for the mass-spring-damper system was verified with the vehicle dynamics team, and the suspension forces obtained through K&C test data from the same team were validated by testing the integrated model as well.

This model is a good way of initiating the suspension system simulation effort, and adding the effects of vertical dynamics simulated through this model to the lateral and longitudinal dynamics will help increase the accuracy of the full-vehicle model. Additionally, integrating the effects of suspension geometry will improve the flexibility in the simulation model.

Quantifying the impact of this project on the overall organization in terms of money is hard, but this effort allows in reducing the operating time of HIL test benches, which helps save both money and time.

7.0 References

1. Tyre and Vehicle Dynamics, Hans Pacejka,
<https://www.sciencedirect.com/book/9780080970165/tire-and-vehicle-dynamics>
2. Race Car Vehicle Dynamics, Milliken & Milliken,
<https://www.sae.org/publications/books/content/r-146/>
3. Automotive Control Systems for Engine, Driveline and Vehicle, U Kiencke, L Nielsen,
<https://www.sciencedirect.com/topics/engineering/automotive-control-system>